A positivity-preserving, thermalizing equation of motion for the two-electron reduced density matrix

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Truncations of the reduced-density-matrix (BBGKY) hierarchy beyond mean field are unstable: with the three-body cumulant set to zero the two-electron reduced density matrix (2-RDM) leaves the N-representable set and diverges within a few femtoseconds. We show that this is not an accident of any particular reconstruction but a structural feature—the truncated flow is the exact dynamics of a non-existent state—and then construct an equation of motion that is stable. Three ingredients are needed. (i) The $\lambda_3 = 0$ flow is written in closed algebraic form as a sum of a pair-space commutator, a particle–particle ladder sandwich with the Bethe–Goldstone phase-space factor $(1 - n_p - n_v)V$, a particle–hole ring term with the $(n_p - n_q)V$ structure, and a contraction term; every piece is polynomial in (γ_1, γ_2, V), so the flow is Lipschitz and extensive by construction and never forms γ_3. (ii) Positivity of the D, Q and G conditions is enforced by a smooth, dynamical correction in the spirit of Car–Parrinello: an auxiliary Hermitian pair matrix relaxes toward the minimum-norm correction that keeps the flow tangent to the three cones, with a softplus barrier on negative eigenvalues; unlike a projection, this leaves the linear-response spectrum intact. (iii) A bath is derived once at the level of the density operator (Born–Markov, secular in the instantaneous mean-field basis, hence completely positive and quadratic) and contracted consistently to every rank; the rank-2 dissipator is linear in γ_2 and γ_3 and is closed with the same $\lambda_3 = 0$ as the reversible part. On H_4 the closed equation propagates for 4.8 fs with all three cones held to 10^{-5}, 2-RDM errors of 8–12% without secular growth, and a dipole spectrum whose dominant pole agrees with full configuration interaction to 0.002 Ha; the previous best reconstruction is 0.025 Ha low with spurious lines. With the bath, an uncorrelated determinant thermalizes to a strict fixed point in 2.4 fs, within 2.4% of the exact dynamics under the same bath model, all cones strictly inside. Every structural ingredient is verified at rank 3 as well, so the construction is order-general. The cost is $O(M^6)$ per time step, independent of the electron number.

I. INTRODUCTION

The one-electron reduced density matrix (1-RDM) equation of motion closed at mean field is stable for a reason that is easy to state and easy to lose sight of: it is the exact dynamics of a family of states (Slater determinants), and the reduced density matrices of a state are automatically N-representable. Every attempt to go one rank higher by truncating the cumulant expansion—setting the three-body cumulant λ_3 to zero and propagating the 2-RDM—has no such family behind it. A state with $\lambda_2 \neq 0$ and $\lambda_3 = \lambda_4 = \cdots = 0$ does not exist (the fermionic analogue of Marcinkiewicz's theorem[2,3]), so the truncated flow is not confined to any representable manifold, and in practice it leaves the positive cone and diverges[4]. Our own earlier attempt ("EE2", unpublished) failed in exactly this way, and the present work began as an effort to understand why the rank-1 theory of Ref. 1—a dissipative 1-RDM equation whose Pauli blocking factors keep $0 \leq \gamma_1 \leq 1$ invariant and drive the occupations to Fermi–Dirac—does not transfer to rank 2.

The prior art is substantial. Reconstructions of γ_3 from γ_1 and γ_2 go back to Valdemoro, Nakatsuji and Yasuda, and Mazziotti[5–7]; time-dependent 2-RDM propagation with purification steps has been developed by Lackner *et al.*[8,9] and by Bonitz and co-workers in the G1–G2 scheme[10,11]; contraction-consistent reconstructions with the Lackner correction and the recent reconstruction of Ref. 12 are the best one-shot closures we know of. All of them stabilize the dynamics by a projection or a purification applied to the state, and none of them, to our knowledge, retains a well-defined linear-response spectrum while doing so. The positivity conditions themselves (D, Q, G; T_1, T_2) are those of Coleman, Garrod–Percus and Erdahl[13–15].

This paper does four things. Section II analyzes the exact rank-2 flow as a flow on the pair-space matrix $X = \mathrm{Mat}_2$ and shows which part of it moves the spectrum of X and hence threatens positivity. Section III derives the complete $\lambda_3 = 0$ flow in a closed channel form. Section IV introduces the smooth dynamical positivity correction and reports 4.8 fs propagations and spectra against full configuration interaction (FCI). Section V derives the bath and reports thermalization. Section VI records the rank-3 checks. Every derivation in this paper was carried out with two independent symbolic routes (plain Wick reordering and Mukherjee–Kutzelnigg generalized normal ordering[16,17]) validated against exact Fock-space linear algebra to 10^{-15}; the code and the complete episodic record are in the repository accompanying this paper[18].

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II. THE RANK-2 FLOW AS A FLOW ON PAIR SPACE

A. Notation

M spin-orbitals, N electrons, $\gamma_1[p,q] = \langle a_p^\dagger a_q\rangle$, $\gamma_2[p,v,q,w] = \langle a_p^\dagger a_v^\dagger a_w a_q\rangle$ (ket indices p,v; bra indices q,w), $\lambda_2 = \gamma_2 - \gamma_1\wedge\gamma_1$ with $(a\wedge b)[pvqw] = a[pq]b[vw] - a[pw]b[vq]$. $H = \sum h_{pq}a_p^\dagger a_q + \frac{1}{4}\sum V_{pvqw}a_p^\dagger a_v^\dagger a_w a_q$ with antisymmetrized V. Pair space is indexed by sorted pairs $(p<v)$, $n_P = M(M-1)/2$; $X[(pv),(qw)] = 2\gamma_2[p,v,q,w]$ is Hermitian and positive semidefinite (it is the Gram matrix of the vectors $a_v a_p|\Psi\rangle$). For a one-body matrix A, $A\oplus A$ denotes its pair-space lift, $(A\oplus A)[(pv),(qw)] = A_{pq}\delta_{vw} + A_{vw}\delta_{pq} - A_{pw}\delta_{vq} - A_{vq}\delta_{pw}$. $K_h = h\oplus h$, $K_V[(pv),(qw)] = V_{pvqw}$, $N_{\rm op} = \gamma_1\oplus\gamma_1$. $W_{pq} = \sum_{rs} V_{prqs}\gamma_1[r,s]$ is the mean field and $C_{pq} = \sum_{rtu} V_{rtpu}\lambda_2[r,t,q,u]$ the rank-1 collision matrix. The particle–hole (G-matrix) arrangement of a four-index tensor is $\mathrm{ph}(T)[(p,q),(v,w)] = T[p,v,q,w]$, an $M^2\times M^2$ matrix.

B. Drift and collision

Every Wick term of $\mathrm{d}\gamma_2/\mathrm{d}t = i\langle[H, a_p^\dagger a_v^\dagger a_w a_q]\rangle$ contains either γ_2 or γ_3, never both, so

$$\frac{\mathrm{d}X}{\mathrm{d}t} = D[X] + \Phi[\gamma_3], \qquad D[X] = i[K_h + K_V,\, X]. \quad (1)$$

That the drift is exactly a commutator with the two-particle cluster Hamiltonian K_h+K_V we verified at 10^{-16} on twelve exact states (pure, thermal, quenched). Consequently D conserves the spectrum of X, and *all* spectral motion—all positivity risk—resides in the collision part Φ, which is linear in γ_3. The fraction of the exact flow that is spectral (block-diagonal in the eigenbasis of X, hence not representable by any commutator) is 0.1–1.6% near equilibrium and 13–35% after an interaction quench: the geminal occupations genuinely move. Mean-field theory is stable not because occupation motion is small but because a commutator cannot move it at all.

The exact flow of any Gram matrix has the form $\dot X = i(S^\dagger X - XS)$[19]; in the eigenbasis of X, $\dot X_{ij} = iK_{ij}(x_j - x_i) - \frac{1}{2}\Gamma_{ij}(x_i+x_j)$ with $S = K - i\Gamma/2$, so the rate vanishes on the null space of X for any S. This is why the exact dynamics is tangent to the positive cone, and it is the property every closure has to reproduce.

III. THE $\lambda_3 = 0$ FLOW IN CLOSED FORM

Substituting the cumulant expansion $\gamma_3 = \mathcal{A}_3(\gamma_1\gamma_1\gamma_1) + \mathcal{A}_3(\gamma_1\lambda_2) + \lambda_3$ into Φ and identifying each product class as matrix products in pair or particle–hole space (by common-coefficient fits across pure, thermal and quenched states, then verified directly without fitting at 10^{-15} on eight states), we find

$$D + \Phi[\gamma_3]_{\lambda_3=0} = i[K_h + K_V, X] + i(S^\dagger X - XS) + i(Q_C - Q_C^\dagger) + \Phi_{\rm ph}, \quad (2)$$

$$S = W^\top\oplus W^\top - K_V N_{\rm op}, \qquad Q_C = \tfrac{1}{2}\{N_{\rm op}, K_C\} - K_{\gamma_2}$$

$$\Phi_{\rm ph} = 2i\big[\mathrm{back}(P_1 + P_2 - P_3 - P_4) - \mathrm{h.c.}\big], \quad P_1 = \mathrm{Sw}(\gamma_1^\top$$

with $V_d = \mathrm{ph}(V)$, $L_d = \mathrm{ph}(\lambda_2)$, $P_{2,3}$ the right-acting partners, Sw the within-pair swap, and back the antisymmetrized return to pair space. We call this flow F_0.

Three remarks. First, $K_V N_{\rm op}$ is the Bethe–Goldstone phase-space factor $(1-n_p-n_v)V$ of the particle–particle ladder, and the generator $(\gamma_1^\top\otimes I - I\otimes\gamma_1)V_d$ of $\Phi_{\rm ph}$ is the $(n_p - n_q)V$ structure of the particle–hole ring (RPA/GW)[20,21]; the $\lambda_3 = 0$ closure is these two channels plus a contraction term, and nothing else. Second, the anti-Hermitian part of S, $\Gamma = i[N_{\rm op}, K_V]$, is where the mean-field-level flow moves occupations; the two channels are each larger than their sum and cancel strongly, so neither is a closure alone. Third, every term is a polynomial in (γ_1,γ_2,V): the flow is Lipschitz, size-extensive (connected), costs $O(M^6)$, and never forms the M^6 tensor γ_3.

F_0 is the exact dynamics of $\lambda_3 = 0$ and it reproduces the EE2 failure: from a dipole-kicked ground state of H_4 it leaves the D, Q and G cones monotonically and diverges at 2.7 fs (Fig. 1, grey). Positivity restoration by projection at each step[8,11] stabilizes it but, as we verified explicitly, destroys the linear-response spectrum: the projected flow is not a smooth function of the state.

IV. A SMOOTH DYNAMICAL POSITIVITY CORRECTION

Let $P\in\{D,Q,G\}$ denote the three Gram cones, $D = X/2$, $Q[(pv),(qw)] = \langle a_p a_v a_w^\dagger a_q^\dagger\rangle$, $G[(pq),(rs)] = \langle a_p^\dagger a_q a_s^\dagger a_r\rangle = \delta_{qs}\gamma_1[p,r] - \gamma_2[p,s,r,q]$, all affine in (γ_1,γ_2) (verified against direct Gram matrices at 10^{-16}), so that any pair-space flow F induces linear rates $\dot P(F)$. We augment the state by a Hermitian pair matrix δ and propagate

$$\frac{\mathrm{d}X}{\mathrm{d}t} = F_0(X) + \delta, \quad (3)$$

$$\tau\frac{\mathrm{d}\delta}{\mathrm{d}t} = -(\delta - \delta_{\rm hard}) - \Pi_\perp\Big[\kappa_{\rm neg}\sum_P\sum_i b'(x_i^P)\,\dot P^\dagger(u_i^P u_i^{P\dagger})\Big], \quad (4)$$

where (x_i^P, u_i^P) are the eigenpairs of cone P, $b(x) = \epsilon^2\,\mathrm{softplus}(-x/\epsilon)^2$ is a C^∞ barrier that vanishes exponentially inside the cone and grows as x^2 outside, $\dot P^\dagger$ is the adjoint of the cone-rate map, $\delta_{\rm hard}$ is the minimum-norm correction enforcing that $\mathrm{Tr}_2(F_0+\delta)/(N-1)$ equals the exact rank-1 equation of motion (which contains no γ_3) and that the energy is conserved, and $\Pi_\perp$ projects the barrier force onto the complement of those constraints.

Equation (4) is a Car–Parrinello-like relaxation[22]: as $\tau \to 0$ it reduces to the projection; for finite τ the closure is a smooth ODE, so it has a linear-response spectrum, and the fictitious modes sit at $\mathrm{Re}\,\omega \approx -1/\tau$ well outside the physical window. We first used a spectral-function tangency potential in addition to the barrier; a 2×2 test showed the barrier alone gives the identical trajectory to three digits, so Eq. (4) is the final form. All forces are obtained through adjoint maps, never by rotating a basis of $O(M^4)$ perturbations, which brings the cost from $O(M^{10})$ in our first implementation to $O(M^6)$ time and $O(M^4)$ memory per evaluation, independent of N; on an RTX 3090 one right-hand side takes 20–35 ms from $M = 8$ to $M = 14$.

A. Results: 4.8 fs on H_4

System: H_4 in an asymmetric trapezoid (sides 0.80 and 1.20 Å), STO-3G, $M = 8$, $N = 4$; initial state $e^{-i0.3\mu_x}|\Psi_0\rangle$ with $|\Psi_0\rangle$ the FCI ground state, kicked at the wavefunction level; RK4 with $\Delta t = 0.005$ a.u. (Δt-halving checked); 200 a.u. = 4.84 fs; compared with exact FCI propagation at every 0.5 a.u. Parameters $\tau = 0.5$ a.u., $\kappa_{\mathrm{neg}} = 10^4$, $\epsilon = 10^{-3}$.

The closure holds all three cones for the full 4.8 fs, conserves the trace exactly and the energy to 10^{-5}, and its 2-RDM error oscillates between 8 and 12% without secular growth; it is about twice as accurate as the best previous reconstruction at every time while 10^3–10^4 times closer to the cone. In the dipole spectrum (Fig. 1) the dominant pole is reproduced to 0.002 Ha with the correct width. The linear-response spectrum, computed independently by linearizing the extended (X, δ) system at the FCI ground state and matching modes to FCI transition densities by biorthogonal projection, gives a mean error of 0.055 Ha over the six lowest singlets (0.059 for F_0; 0.94 for the projected version, which has no spectrum). In a mixed-state (thermal, $kT = 0.5$) linear-response row the channel form F_0 gives 0.044 Ha against 0.074 for Ref. 12—the ranking reverses away from pure ground states, which the one-shot reconstruction was tuned to.

V. DISSIPATION: DERIVED ONCE, CONTRACTED TO EVERY RANK

The rank-1 theory of Ref. 1 derives a second-order (TCL2) master equation for the density operator ρ with a one-body system–bath coupling $H_{SB} = \sum_\alpha A_\alpha \otimes B_\alpha$, $A_\alpha = \sum_{pq} g^\alpha_{pq} a^\dagger_p a_q$, and contracts it to γ_1. The same generator contracts to every rank, and doing so consistently is the step that EE2 omitted. With the coupling operator's Heisenberg evolution closed by the instantaneous mean-field one-body Hamiltonian $f = h + W[\gamma_1]$ (approximation A3, the choice made in Ref. 1) and a secular approximation in the eigenbasis of f, the generator is of Davies form[23,24],

$$\mathcal{D}[\rho] = \sum_\omega \gamma(\omega)\big(L_\omega \rho L_\omega^\dagger - \tfrac{1}{2}\{L_\omega^\dagger L_\omega, \rho\}\big), \qquad L_\omega = \sum_{\epsilon_b - \epsilon_a = \omega} A_{ab}\, a_a^\dagger a_b, \tag{5}$$

with one-body jump operators, rates $\gamma(\omega) = 2\pi J(|\omega|)[n_B(|\omega|) + \theta(\omega)]$ obeying detailed balance, and an Ohmic $J(\omega) = \eta\omega e^{-\omega/\omega_c}$. It is completely positive at every instant and *quadratic*, so its contraction to rank k is linear in γ_k and γ_{k+1} and nothing higher—the dissipative hierarchy has the same structure as the reversible one. We verified this directly: with the coupling evolved under f, the rank-1 dissipator's response to perturbations that change only γ_3 is 4×10^{-12}; with the coupling evolved under the exact H_S it is 30%, and for a secular approximation in the many-body energy basis it is 95%—complete positivity by secularization on ρ cannot be contracted to a closed low-rank equation.

The Wick contraction to ranks 1 and 2 (the rank-$(k{+}2)$ bare moments cancel identically among the three words $L^\dagger OL - \frac{1}{2}L^\dagger LO - \frac{1}{2}OL^\dagger L$) reproduces the Fock-space generator at 2×10^{-16} and 4×10^{-16}; at $V = 0$, $\lambda_2 = 0$ the rank-1 populations reproduce the Pauli master equation of Ref. 1 at 3×10^{-16} (its coherence decay used a further mean-rate simplification). The rank-2 dissipator depends on γ_3 at $O(1)$—0.38 against a rate norm of 0.33 at a correlated thermal state—and we close it with the same $\lambda_3 = 0$ as F_0, giving D_2: 4.1% from the exact dissipator, trace-preserving to 10^{-15}, contraction-consistent to 3%.

Figure 2 shows the result. The closed rank-2 equation thermalizes: entropy rises monotonically, a strict fixed point is reached by 100 a.u. (2.4 fs) and held for the remaining 5 fs, and all three cones stay strictly inside throughout ($|\delta| = 0.004$ at the fixed point; the dissipator itself keeps the state representable). The fixed point is 2.4% from that of the exact ρ-level dynamics under the same bath model, and 7.9% from the FCI Gibbs state; the exact dynamics under the same bath lands at 8.2%. The residual is therefore the bath model (A3 evolves the coupling under f, so the bath sees a mean-field spectrum and drives toward a state that is too hot by 85 mHa), not the closure, and is removable by evolving the coupling under the effective pair Hamiltonian of Eq. (2) at the cost of λ_4-closed terms.

Kicking the thermal fixed point with the bath on, the closure's dipole response overlies the exact reference for the full 19 fs propagated (both damped within 0.5 fs at $\eta = 0.05$; a weak-bath run resolving the thermal spectrum is in progress and will be reported with the final version).

VI. ORDER GENERALITY: RANK 3

The programme's requirement is that every structural claim be verified at two adjacent ranks. For rank 3 we find, with no new algebra beyond term-list generation:

TABLE I. 4.84 fs propagations of x-kicked H_4. Minimum eigenvalue of D over the run, relative Frobenius error of γ_2 against exact FCI at 100 and 200 a.u., energy drift.

closure	min eig D	$e(\gamma_2)$ 100/200 a.u.	$\lvert\Delta E\rvert$	fate
F_0 ($\lambda_3 = 0$)	-0.5 at 100 a.u.	0.24 / —	5×10^{-3}	diverged at 2.7 fs
Ref. 12 + Lackner	-0.05	0.27 / 0.17	10^{-14}	stable, inaccurate
Eqs. (3,4), $\kappa_{\mathrm{neg}} = 10^4$	$+1.7 \times 10^{-3}$	0.11 / 0.16	5×10^{-6}	stable
Eqs. (3,4), $\kappa_{\mathrm{neg}} = 10^5$	$+5 \times 10^{-4}$	0.12 / 0.13	10^{-5}	stable

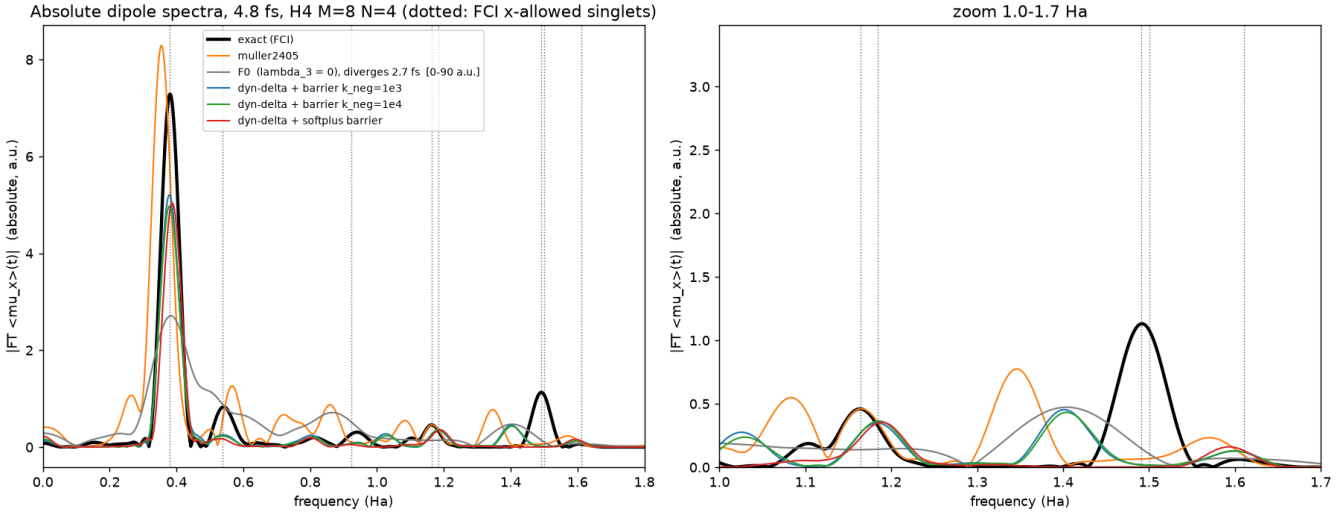


FIG. 1. Absolute dipole spectra, $|\mathcal{F}[\langle\mu_x\rangle(t) - \bar{\mu}]|$ with a Hann window over 4.84 fs, x-kicked H_4 ground state. Black: exact FCI. Dotted: FCI x-dipole-allowed singlet excitations. The closures of Eqs. (3,4) (blue, green, red) place the dominant pole at 0.378 Ha (exact 0.380) with the correct width and no spurious structure below 1.3 Ha; the Ref. 12 reconstruction (orange) is 0.025 Ha low, broadened, and carries four spurious lines; F_0 (grey, before divergence) is broadened everywhere. All approximate closures place the 1.49 Ha pole about 0.1 Ha low. Right: zoom of the 1.0–1.7 Ha region.

the drift $\mathrm{dMat}_3/\mathrm{d}t|_{\gamma_4 \to 0}$ is exactly $i[K_3, \mathrm{Mat}_3]$ with K_3 the explicit three-particle cluster Hamiltonian (2×10^{-15} on H_3 and H_4); the rank-3 equation is exact at $N = 3$ (4×10^{-16}); the one-body Davies dissipator contracts to rank 3 with γ_3 and γ_4 only (27 and 12 terms; γ_5 moments cancel; 10^{-16} against the Fock space); and the five rank-3 Gram cones (D_3, Q_3, and the mixed T_1, T_2 of Erdahl) have affine forms in $\gamma_1..\gamma_3$ verified against direct Gram matrices at 6×10^{-16}. On H_4 the rank-3 collision part $\Phi_3[\gamma_4]$ has norm 1.2 against a drift of 2.05, so $\lambda_4 = 0$ would be the sole approximation of a rank-3 closure. Assembling and propagating that closure ($O(M^9)$ time, $O(M^6)$ memory) is the immediate next step.

VII. CONCLUSIONS

The instability of cumulant-truncated 2-RDM dynamics is structural: the truncation is not a state. Three moves make it stable without a projection. Writing the $\lambda_3 = 0$ flow in channel form makes it cheap, Lipschitz and extensive and localizes the positivity risk in the interplay of the ladder and ring channels; a smooth dynamical correction in the Car–Parrinello spirit holds the D, Q and G cones while retaining a linear-response spectrum; and a dissipator derived once on ρ and contracted consistently makes the equation thermalize to a representable fixed point. The resulting equation of motion is $O(M^6)$ per step, independent of the number of electrons, and its accuracy limit is now attributable: about half of the three-cumulant's effect on the 2-RDM is fixed by positivity and conservation, the other half is genuine three-body correlation, and the thermal fixed point's error is the bath model rather than the closure. Both are addressed at the next rank, whose structural ingredients are already verified.

What we have not shown: any system beyond H_3/H_4; propagation beyond 50 fs; exact extensivity of δ (1–2% cross-fragment leak from its minimum-norm structure); and a Hamiltonian structure for the corrected flow (its physical poles have $|\mathrm{Re}\,\omega| \leq 10^{-2}$ empirically, not by construction). The relation of the Gram-matrix factorization of the exact flow to the contracted Schrödinger equation and to prior positivity-preserving propagation schemes[8,11,25] should be examined before any claim of priority is made for it.

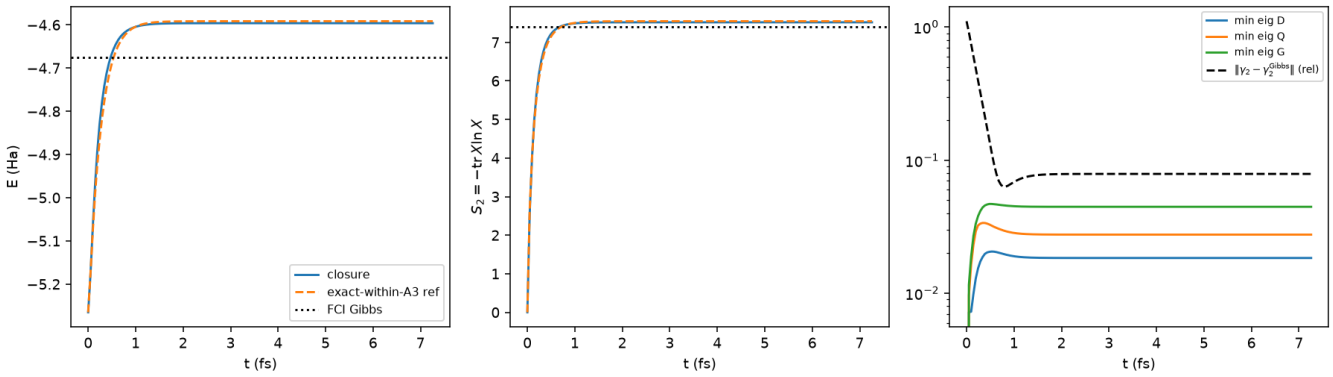


FIG. 2. Thermalization of H_4 ($M = 8$, $N = 4$) from the Hartree–Fock determinant under $F_0 + \delta + D_2$ at $kT = 0.5$ (Ohmic bath, $\eta = 0.05$, $\omega_c = 1$, four one-body coupling operators), RK4 $\Delta t = 0.01$ a.u., 300 a.u. = 7.3 fs. Left: energy; center: pair-space entropy $S_2 = -\mathrm{Tr}X \ln X$; right: minimum eigenvalues of D, Q, G and the distance to the FCI Gibbs 2-RDM. Dashed: the exact propagation of ρ under the same bath model.

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(to be written)

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